

Q1) in order to show: $N(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \leq N(\Theta, d, \frac{\varepsilon}{C})$

we will show that every $\frac{\varepsilon}{C}$ -cover on Θ is an

ε -cover on $\mathcal{F}$. take $A = \{\theta_1, \theta_2, \dots, \theta_m\}$ to be an

$\frac{\varepsilon}{C}$ -cover on Θ , take $B := \{f_\theta : \theta \in A\}$, we'll show that this is an ε -cover on $\mathcal{F}$. note that we have;

$$\forall x \in \mathcal{X} : |f_\theta(x) - f_{\theta'}(x)| < C d(\theta, \theta')$$

$$\|f_\theta - f_{\theta'}\|_\infty = \sup_{x \in \mathcal{X}} |f_\theta(x) - f_{\theta'}(x)| \rightarrow \text{then we have for } x^* = \arg\max(\dots)$$

$$\text{therefore we have } |f_\theta(x^*) - f_{\theta'}(x^*)| = \|f_\theta - f_{\theta'}\|_\infty < C d(\theta, \theta')$$

take an arbitrary $f_\theta \in \mathcal{F}$.

there exists a $\theta_c \in \Theta$ such that $d(\theta, \theta_c) < \frac{\varepsilon}{C}$

so we have:

$$\|f_\theta - f_{\theta_c}\| \leq C d(\theta, \theta_c) \leq \varepsilon.$$

therefore we have if A is an $\frac{\varepsilon}{C}$ -cover on Θ ,

subsequently there exists an ε -cover B on $\mathcal{F}$ with the

same Cardinality therefore: $N(\mathcal{F}, \|\cdot\|_\infty, \varepsilon)$

$$\leq N(\Theta, d, \frac{\varepsilon}{C}) \quad \square.$$

Q2) the chaining argument can be written as follows:

$$\mathbb{E}[\sup_{t \in T} X_t] \leq \mathbb{E}[X_{t_0}] + \sum_{k=1}^n \mathbb{E}[\sup_{t \in T} \{X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\}] \\ + \mathbb{E}[\sup_{t \in T} \{X_t - X_{\pi_n(t)}\}].$$

$\mathbb{E}[X_{t_0}] = 0$, since X_t is subgaussian then $X_{\pi_k(t)} - X_{\pi_{k-1}(t)}$ is subgaussian with variance proxy $(d(\pi_k(t), \pi_{k-1}(t))^2)$

and we have: $d(\pi_k(t), \pi_{k-1}(t)) \leq d(\pi_k(t), t) + d(\pi_{k-1}(t), t) \leq 3 \times 2^{-k}$

then by $\mathbb{E}[\sup X_t] \leq \sqrt{2\sigma^2 \lg |T|}$

we have $\sum_{k=1}^n \mathbb{E}[\sup \{X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\}] \leq \sum_{k=1}^n 6 \times 2^{-k} \sqrt{\lg(N_k)}$

now we'll bound this sum by an integral

$$\sum_{k=1}^n (\dots) \leq \sum_{k=1}^n (\dots) = 2 \sum_{k=1}^n \int_{2^{-k-1}}^{2^{-k}} \sqrt{\lg(N(\tau, d, 2^{-k}))} d\varepsilon$$

since the integrand is decreasing with increase of radius we can write

$$\leq 2 \sum_{k=1}^n \int_{2^{-k-1}}^{2^{-k}} \sqrt{\lg(N(\tau, d, \varepsilon))} d\varepsilon = 2 \int_{2^{-n-1}}^{\infty} \sqrt{\lg(N(\tau, d, \varepsilon))} d\varepsilon$$

Q2) continued:

because of the Lipschitz property we have:

$$\forall t \in T : |X_t - X_{n(t)}| \leq C \times 2^{-n}$$

$$\therefore \sup \{ \dots \} \leq C \times 2^{-n} \rightarrow \mathbb{E} \sup \{ X_t - X_{n(t)} \} \leq \mathbb{E}[C] \times 2^{-n}$$

$$\text{now set } 2^{-n-1} = \delta$$

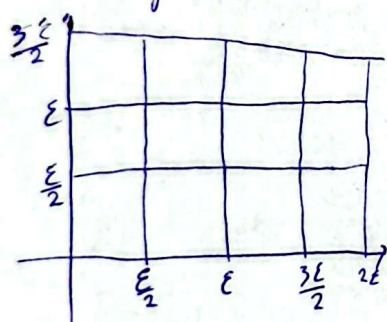
therefore we have:

$$\mathbb{E} \sup X_t \leq 2\delta \mathbb{E}[C] + 12 \int_{\delta}^{\infty} \sqrt{\log(K(\tau, \delta, \epsilon))} d\epsilon$$

taking an infimum on the RHS yields the desired result. $\square$

Q3) we'll take a similar route to the 1D case, first we'll show the inequality for the 2D case and Generalize for higher dimensional cases.

Imagine a 2D Grid like so, we'll treat this Grid like the domain for our function f , since our function is 1-Lipschitz therefore by going from a square to a nearby square:



(because I_{x_1}, I_{x_2} are neighbours)

$$\mathbb{P}_2 \quad d \left(\min_{x \in I_{x_1}} (f(x)), \min_{x \in I_{x_2}} (f(x)) \right) \leq \epsilon$$

3.1) continued:

and also in a square we have:

$$d\left(\max_{x \in I_k} (f(x)), \min_{x \in I_k} (f(x))\right) \leq \frac{\varepsilon}{2} \quad (\text{note that these arguments are also applicable in the General case not just 2D})$$

partition $[0,1] \rightarrow J_i : [c(i-1), c(i)]$

for an arbitrary $f \in F$, define $\pi(f)(x) = \frac{\max J_i + \min J_i}{2}$

when $x \in I_k$, $\min f(I_k) \in J_i$. (I_k 's represent the d -dimensional squares) (with width $\frac{\varepsilon}{2}$)

$$\|\pi(f)(x) - f(x)\|_\infty \leq \varepsilon \quad \left(\text{note that because of } \mathcal{A}_1 \right. \\ \left. \text{even if } \min_{x \in I_k} f(x) = \max J_i - \delta \right.$$

$$\text{then } \max f(I_k) \leq \max J_i + \frac{\varepsilon}{2} - \delta$$

$$\text{and } \max_{x \in I_k} (f(x) - \pi(f)(x)) \leq \varepsilon - \delta \quad \left. \right)$$

the above argument shows that the set of all possible $\pi(f)$ covers F_0 . therefore the only thing

left to do is to bound the Cardinality of all possible

$\pi(f)$, because of $(\mathcal{A}_2)$ we know that in transition between two blocks the (min) changes at most (ε) therefore in one block we're at $\frac{(2K+1)\varepsilon}{2}$ at the next one (meaning a neighbour) we can go to $\frac{2K+1}{2}(\varepsilon) \pm 1$

or stay at $(\frac{2K+1}{2})\varepsilon$, meaning in each step of

the way we have 3 options when transitioning

3.1) continued:

therefore to count all the possible $\pi(f)$'s, we have to start at I_0 and step by step go to all the neighbors since $f_{(0)} = 0$, we'll set $\pi(f)_{(0)} = 0 \quad \forall f \in \mathcal{F}$

note that the cardinality of the boxes set $\{I_k\}$ is $(\lceil \frac{2}{\epsilon} \rceil)^d$. therefore all the combinations

are at the order of: $N(\mathcal{F}, \|\cdot\|_\infty, c) \leq 3^{(\lceil \frac{2}{\epsilon} \rceil)^d} \leq \exp(-\frac{c}{\epsilon^d})$
 $c \approx 2^d$. this concludes the proof.

3.2) to show that we can apply the Dudley bound to this stochastic process, we first have to show it's sub-gaussian with respect to some meter.

$$Z_f = \sum \frac{f(x_k) - \mu_f}{n} \rightarrow |Z_f - Z_g| = \frac{\sum (f(x_k) - g(x_k)) - (\mu_f - \mu_g)}{n}$$

$$|f(x_k) - g(x_k)| \leq \|f - g\|_\infty \rightarrow |Z_f - Z_g| \sim \text{sub-gaussian} \left(\frac{\|f - g\|_\infty^2}{n} \right)$$

therefore the above process is sub-gaussian w.r.t.

$\frac{\|\cdot\|_\infty}{\sqrt{n}}$ as the distance.

3.2) continued:

therefore we have:

$$\begin{aligned} E[W_1(\mu_n, \mu)] &\leq 12 \int_0^\infty \sqrt{\log(K(F, \frac{\|\cdot\|_\infty}{\sqrt{n}}, \varepsilon))} d\varepsilon \\ &= 12 \int_0^\infty \sqrt{\log(K(F, \|\cdot\|_\infty, \varepsilon\sqrt{n}))} d\varepsilon = 12 \frac{1}{\sqrt{n}} \int_0^\infty \sqrt{\log(K(F, \|\cdot\|_\infty, \varepsilon))} d\varepsilon \\ &\leq \frac{12}{\sqrt{n}} \int_0^\infty \sqrt{\log(e^{\frac{C}{\varepsilon^d}})} d\varepsilon = \frac{12}{\sqrt{n}} \int_0^\infty \frac{\sqrt{C}}{\varepsilon^{\frac{d}{2}}} d\varepsilon \end{aligned}$$

the above integral diverges for $d \geq 2$

3.3) first let's calculate the Lipschitz Constant of the process

$$\begin{aligned} |Z_f - Z_g| &= \left| \frac{\sum (f(x_k) - g(x_k)) - (\mu f - \mu g)}{n} \right| \leq \frac{\sum |f(x_k) - g(x_k)| + |\mu f - \mu g|}{n} \\ &\leq \frac{\sum (1 + 1) \|f - g\|_\infty}{n} = 2 \|f - g\|_\infty \rightarrow C = 2\sqrt{n} \quad (\text{in norm: } \frac{1}{\sqrt{n}} \|f - g\|_\infty) \end{aligned}$$

therefore by (Q2) we have:

$$E[W_1(\mu, \mu_n)] \leq \inf_{\delta_1} \left\{ 2\delta_1 \times 2\sqrt{n} + 12 \int_{\delta_1}^\infty \sqrt{\log(K(F, \frac{\|\cdot\|_\infty}{\sqrt{n}}, \varepsilon))} d\varepsilon \right\}$$

from the argument in 3.2 the above transforms to:

$$\leq \inf_{\delta_1} \left\{ 4\delta_1 \sqrt{n} + \frac{\tilde{C}}{\sqrt{n}} \int_{\delta_1 \sqrt{n}}^{\frac{1}{2}} \frac{1}{\varepsilon^{\frac{d}{2}}} d\varepsilon \right\}$$

3.3) continued: since: $\inf_{\delta > 0} (f(\delta\sqrt{n})) = \inf_{\delta' > 0} (f(\delta'))$ set: $\delta\sqrt{n} = \delta'$ and minimize on it

$$d=1 \rightarrow \inf \left\{ 4\delta + \frac{\tilde{C}}{\sqrt{n}} \int_{\delta}^{\frac{1}{2}} \frac{1}{\sqrt{\varepsilon}} d\varepsilon \right\}$$

$$= 4\delta + \frac{2\tilde{C}}{\sqrt{n}} (\sqrt{\frac{1}{2}} - \sqrt{\delta})$$

$$\rightarrow \frac{d}{d\delta}(\dots) = 4 + \left(\frac{2\tilde{C}}{\sqrt{n}}\right)\left(-\frac{1}{2\sqrt{\delta}}\right) = 0 \rightarrow \sqrt{\delta} = \frac{\tilde{C}}{4\sqrt{n}}$$

$$\rightarrow \leq \frac{4\tilde{C}^2}{16n} + \frac{\sqrt{2}\tilde{C}}{\sqrt{n}} - \frac{\tilde{C}^2}{2n} \lesssim n^{-\frac{1}{2}} \quad (\text{the dominate term is } \frac{1}{\sqrt{n}})$$

$$d=2 \rightarrow \inf \left\{ 4\delta + \frac{\tilde{C}}{\sqrt{n}} \int_{\delta}^{\frac{1}{2}} \frac{1}{\varepsilon} d\varepsilon \right\}$$

$$= 4\delta + \frac{\tilde{C}}{\sqrt{n}} (\ln(\frac{1}{2}) - \ln(\delta))$$

$$\frac{d}{d\delta}(\dots) = 4 + \frac{\tilde{C}}{\sqrt{n}} \frac{-1}{\delta} = 0 \rightarrow \delta = \frac{\tilde{C}}{4\sqrt{n}}$$

$$\rightarrow \leq \frac{4\tilde{C}}{\sqrt{n}} + \frac{\tilde{C}}{\sqrt{n}} \ln(2) + \frac{\tilde{C}}{2\sqrt{n}} \ln\left(\frac{16n}{\tilde{C}^2}\right) \lesssim n^{-\frac{1}{2}} \log n$$

$$d \geq 3 \quad \inf \left\{ 4\delta + \frac{\tilde{C}}{\sqrt{n}} \int_{\delta}^{\frac{1}{2}} \frac{1}{\varepsilon^{\frac{d}{2}}} d\varepsilon \right\}$$

$$= 4\delta + \frac{\tilde{C}}{\sqrt{n}} \frac{\varepsilon^{-\frac{d}{2}+1}}{-\frac{d}{2}+1} \Big|_{\delta}^{\frac{1}{2}} = 4\delta + \frac{\tilde{C}}{\sqrt{n}} \frac{\left(\left(\frac{1}{2}\right)^{1-\frac{d}{2}} - (\delta)^{1-\frac{d}{2}}\right)}{1-\frac{d}{2}}$$

$$\frac{d}{d\delta}(\dots) = 4 + \frac{\tilde{C}}{\sqrt{n}} \frac{-1}{\delta^{\frac{d}{2}}} = 0 \rightarrow \delta = \frac{(\tilde{C})^{\frac{2}{d}}}{n^{\frac{1}{d}} 4^{\frac{2}{d}}}$$

$$\leq \frac{4\tilde{C}^{\frac{2}{d}}}{n^{\frac{1}{d}} 4^{\frac{2}{d}}} + \alpha \frac{1}{\sqrt{n}} + \beta \frac{1}{n^{\frac{1}{d}}} \lesssim n^{-\frac{1}{d}}$$

□.

3.4) we want to provide a lower bound for $E[\min \|x - X_i\|] \geq n^{-1/d}$

$$E[\min \|x - X_i\|] = \int_0^\infty P(\min \|x - X_i\| > t) dt$$

$$P(\min \|x - X_i\| > t) = P(\|x - X_i\| > t)^n$$

(independence)

note that for $A \subseteq [0, 1]^d$, $P(X \in A) = \int_A \rho(x) dx$

therefore we have: $P(\|x - X_i\| > t) = 1 - \int_{B(x, t)} \rho(x) dx$

$$\int_{B(x, t)} \rho(x) dx \leq \|\rho\|_\infty \nu_d(B(x, t)) = \|\rho\|_\infty \nu_d t^d$$

$$\Rightarrow P(\|x - X_i\| > t) \geq 1 - \underbrace{\|\rho\|_\infty \nu_d}_\alpha t^d$$

$$\Rightarrow E[\min \|x - X_i\|] = \int_0^\infty P(\min \|x - X_i\| > t) dt \geq \int_0^{n^{-1/d}} P(\dots) dt \geq \int_0^{n^{-1/d}} (1 - \alpha t^d)^n dt$$

$$\geq \int_0^{n^{-1/d}} (1 - n\alpha t^d) dt \geq \left(t - \frac{n\alpha}{d+1} t^{d+1} \right) \Big|_0^{n^{-1/d}} = n^{-1/d} - \frac{\alpha}{d+1} n^{-1/d} \geq \frac{1}{2} n^{-1/d}$$

(1 - \alpha)^n \geq 1 - n\alpha

□.

Q 3.5)

$$E[W_1(\mu_n, \mu)] = E\left[\sup_{f \in \mathcal{F}} \{\mu_n f - \mu f\}\right]$$

$$\mathcal{F}_0 : \{f: [0,1]^d \rightarrow \mathbb{R} : f \text{ is 1-Lipschitz}\}$$

consider $f(x) = -\min_i \|x - x_i\|$, note that f is 1-Lipschitz

because $(\forall x \in [0,1]^d) -\min_i \|x - x_i\|_2 = \min_j \|g - x_j\|$

suppose j^* minimizes the second function, then

$$\min_i \|x - x_i\| \leq \|x - x_{j^*}\| \rightarrow -\min_i (\dots) + \min_j (\dots) \leq \|g - x_{j^*}\| - \|x - x_{j^*}\|$$

$$\stackrel{\text{triangle}}{\text{ineq.}} \leq \|x - g\|_2 \leq \|x - g\|_\infty \rightarrow f(x) \text{ is 1-Lipschitz w.r.t. } \ell_\infty \text{ norm.}$$

therefore we have: $E\left[\sup_{f \in \mathcal{F}} \{\mu_n f - \mu f\}\right] \geq E[(\mu_n g - \mu g) | g = f(x)]$

$$\mu_n f = \frac{\sum_{k=1}^n f(x_k)}{n} = \frac{\sum_{k=1}^n -\min_i \|x_k - x_i\|}{n} = 0$$

$$\rightarrow \mu_n f - \mu f = 0 - E[-\min_i \|x - x_i\|]$$

$$\rightarrow E[W_1(\mu_n, \mu)] \geq E[\min_i \|x - x_i\|] \geq n^{-\frac{1}{d}} \quad \square$$

Q 4.1)

$$\begin{aligned} \text{Excess}(r) &= L(\hat{\theta}) - L(\theta^{opt}) = (L(\hat{\theta}) - L_n(\hat{\theta})) + (L_n(\hat{\theta}) - L_n(\theta^{opt})) \\ &\quad + (L_n(\theta^{opt}) - L(\theta^{opt})) \end{aligned}$$

note that $L_n \hat{\theta} \leq L_n \theta^{opt}$ (because $\hat{\theta}$ is its minimizer)

therefore we have: $\text{Excess}(r) \leq (L(\hat{\theta}) - L_n(\hat{\theta})) + (L_n(\theta^{opt}) - L(\theta^{opt}))$

$$\leq |L\hat{\theta} - L_n\hat{\theta}| + |L\theta^{opt} - L_n\theta^{opt}| \leq 2 \sup_{\theta \in \Theta} |L_n(\theta) - L(\theta)|$$

$$4.2) \quad Z_\theta = L_n(\theta) - L(\theta) \quad \mathbb{E}[Z_\theta] = \mathbb{E}_p L_n \theta - \mathbb{E} L_\theta = 0$$

$$L_n(\theta) = \frac{1}{n} \sum_j \ell_\theta(x_j) \quad , \quad \ell_\theta(x) = -\log p_\theta(x) = -\sum_{i=1}^T \log p_\theta(x_i | x_{-i}, c)$$

in order to show that Z_θ is subgaussian we have to show that

$$\mathbb{E} [\exp(\lambda(Z_\theta - Z_{\theta'}))] \leq \exp\left(\frac{\lambda^2 d(\theta, \theta')^2}{2}\right)$$

$$Z_\theta - Z_{\theta'} = (L_n(\theta) - L_n(\theta')) - \mathbb{E}_p [L_n(\theta) - L_n(\theta')]$$

$$L_n(\theta) - L_n(\theta') = \frac{1}{n} \sum_j \ell_\theta(x_j) - \ell_{\theta'}(x_j)$$

4.2) continued:

$$\ell_0(X) - \ell'_0(X) = \sum_1^I \log p_{\theta'}(x_i | x_{-i}, c) - \log p_{\theta}(x_i | x_{-i}, c)$$

we have: $p_{\theta}(x_i | x_{-i}, c) = \text{softmax}(Z_{\theta}(x_{-i}, c))_{ij}$

$$= \frac{e^{Z_i}}{\sum_1^{\tilde{n}} e^{Z_j}} \rightarrow \log p_{\theta}(x_i | \dots) = Z_i - \log(\sum_1^{\tilde{n}} e^{Z_j})$$

$$\rightarrow \log(p_{\theta'}(x_i | \dots)) - \log(p_{\theta}(x_i | \dots)) = Z_i - Z'_i - (\log(\dots) - \log(\dots))$$

note that $|Z_i - Z'_i| \leq \|Z - Z'\|_{\infty} \leq \|Z - Z'\|_2$

we'll show that $\log(\sum e^{Z_i}) - \log(\sum e^{Z'_i}) \leq \|Z - Z'\|_{\infty}$

set $\delta := \|u - v\|_{\infty} \rightarrow u_j \leq v_j + \delta \rightarrow e^{u_j} \leq e^{\delta} e^{v_j}$

$$\rightarrow \sum e^{u_j} \leq e^{\delta} \sum e^{v_j} \quad (\text{by}) \quad \log \sum e^{u_j} \leq \delta + \log \sum e^{v_j}$$

$$\rightarrow \log(\dots) - \log(\dots) \leq \|u - v\|_{\infty}$$

$$\rightarrow Z_i - Z'_i - (LSE(z) - LSE(z')) \leq |Z_i - Z'_i| + |LSE(z) - LSE(z')|$$

$$\leq 2\|Z_i - Z'_i\|_{\infty} \leq 2\|Z_i - Z'_i\|_2 \leq 2L_{z'} \|\theta - \theta'\|_2$$

4.2) continued:

therefore we have bounded the R.V.: $l_\theta(X) - l_{\theta'}(X)$

by the Constant: $\leq 2TL_Z \|\theta - \theta'\|_2$ (sum of T : $\log p_\theta - \log p_{\theta'} \leq 2L_Z \|\theta - \theta'\|_2$)

note that $Z_\theta - Z_{\theta'} = \frac{1}{n} \sum l_\theta(x_i) - l_{\theta'}(x_i) - \mathbb{E}[L_\theta(\theta) - L_{\theta'}(\theta')]$

therefore by Azuma-Hoeffding lemma $Z_\theta - Z_{\theta'}$ is sub-g

with $\sigma^2 = n \left(\frac{1}{2} \cdot \frac{2TL_Z \|\theta - \theta'\|_2}{n} \right)^2 \rightarrow \sigma = \frac{TL_Z}{\sqrt{n}} \|\theta - \theta'\|_2$

(note that if $a < Z < b \rightarrow Z \sim \text{sub-g} - \left(\left(\frac{b-a}{2} \right)^2 \right)$)

4.3)

$$\mathbb{E} \left[\sup_{\theta \in \Theta} |Z_\theta| \right] \leq 12 \int_0^\infty \sqrt{\log \mathcal{N}(\cdot, d, \varepsilon)} d\varepsilon$$

$$d(\theta, \theta') = \frac{TL_Z}{\sqrt{n}} \|\theta - \theta'\|_2 \rightarrow \mathcal{N}(\Theta, d, \varepsilon) = \mathcal{N}(\Theta, \|\cdot\|_2, \frac{\sqrt{n}}{TL_Z} \varepsilon)$$

$$\rightarrow 12 \int_0^\infty \sqrt{\log \mathcal{N}(\dots)} d\varepsilon = 12 \int_{\left(\varepsilon' = \frac{\sqrt{n}}{TL_Z} \varepsilon\right)}^\infty \sqrt{\log \mathcal{N}(\Theta, \|\cdot\|_2, \varepsilon')} \times \frac{TL_Z}{\sqrt{n}} d\varepsilon'$$

$$\Theta = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \leq R\} \quad \mathcal{N}(\Theta, \|\cdot\|_2, \varepsilon) \leq \mathcal{P}(\Theta, \|\cdot\|_2, \varepsilon)$$

volume argument:

$$\mathcal{P} \times \text{Vol} \left(B\left(\frac{\varepsilon}{2}\right) \right) \leq \text{Vol} \left(B\left(R + \frac{\varepsilon}{2}\right) \right) \rightarrow \mathcal{P} \leq \left(\frac{R + \frac{\varepsilon}{2}}{\frac{\varepsilon}{2}} \right)^d$$

4.3) contnued:

$$\mathbb{E} \left[\sup_{\theta \in \Theta} |Z_{\theta}| \right] \leq 12 \frac{T L_z}{\sqrt{n}} \int_0^R \sqrt{\log(N)} d\epsilon \leq 12 \frac{T L_z}{\sqrt{n}} \int_0^R \sqrt{\log(1 + \frac{2R}{\epsilon})} d\epsilon$$

$$12 \frac{T L_z R \sqrt{d}}{\sqrt{n}} \underbrace{\int_0^1 \sqrt{\log(1 + \frac{2}{\epsilon})} d\epsilon}_{\approx 1.35} \approx 16.2 \frac{T L_z R \sqrt{d}}{\sqrt{n}}$$

$$\rightarrow K = 16.2$$

Since $\text{Excess}(P) \leq 2 \sup_{\theta} |L_{\theta}| - L_{\theta^*}$

$$\text{therefore: } \sup \mathbb{E}_P [\text{Excess}(P)] \leq 32.4 \frac{T L_z R \sqrt{d}}{\sqrt{n}}$$

the above bound is linear with T , so doubling T would double the $\mathbb{E}$ value of worst case scenario.
we can increase n to compensate for T .

and no, the number of words in our vocabulary doesn't have anything to do with our Excess bound at least directly.

Q5) partition the horizontal axis with intervals: $I_1, \dots, I_{\lceil \frac{1}{\varepsilon} \rceil}$ with size $\frac{\varepsilon}{2}$

do the same for the vertical axis: $J_1, \dots, J_{\lceil \frac{1}{\varepsilon} \rceil}$, with size $\frac{1}{\varepsilon}$

define: $\pi(f)(u) = \frac{\max J_i + \min J_i}{2}$ whenever: $x \in I_k, f(\min I_k) \in J_i$

now if f is α -lipschitz ($\alpha \leq 1$)

$$|f(x) - \pi f(x)| \leq |f(x) - f(\min I_k)| + |f(\min I_k) - \frac{\max J_i + \min J_i}{2}|$$

$$\leq \alpha |x - \min I_k| + \frac{\max J_i - \min J_i}{2} \leq (1 + \alpha) \frac{\varepsilon}{2} \leq \varepsilon$$

for $\forall f \in F$ we can build a corresponding $\pi(f)$

which is distance wise ε far from f

a naive way to bound the number of $\pi(f)$ s

would be $\lceil \frac{1}{\varepsilon} \rceil^{\lceil \frac{1}{\varepsilon} \rceil} \leq \left(\frac{2}{\varepsilon}\right)^{\lceil \frac{1}{\varepsilon} \rceil}$ (i.e. $\#$ of all possible positions for every I_k)

which satisfies our problems desired bound

but if we go one step further and we'd use the lipschitz property we can achieve

$$N(F, \|\cdot\|_\infty, \varepsilon) \leq \left\lceil \frac{1}{\varepsilon} \right\rceil^{\lceil \frac{1}{\varepsilon} \rceil - 1} \leq \left(\frac{2}{\varepsilon}\right)^{\frac{2}{\varepsilon}}$$

Q6) for this Question we'll use matrix Concentration inequality, particularly (Matrix Chernoff)

since Matrix $A \in \mathbb{R}^{m \times n}$ is a Rectangular matrix, we'll use the dilated version of A to bound its singular values.

$$\mathcal{D}(A) = \begin{pmatrix} 0 & A \\ A^T & 0 \end{pmatrix}_{(m+n) \times (m+n)} \quad \text{[note that } \|A\|_{op} = \|\mathcal{D}(A)\|_{op}]$$

to bound the MGF of A and benefit from the sub-gaussian Hoeffding, we'll decompose $\mathcal{D}(A)$.

$$\mathcal{D}(A) = \sum_{i,j} A_{ij} B_{ij} = \sum_{i,j} x_{ij}, \quad B_{ij} = \begin{pmatrix} 0 & e_i e_j^T \\ e_j e_i^T & 0 \end{pmatrix}$$

Let's try and Bound the mfg of x_{ij}

$$\mathbb{E}[e^{\lambda x_{ij}}] \Rightarrow B_{ij} = U \text{diag}(b_1, \dots, b_d) U^T \rightarrow e^{\lambda x_{ij}} = U \text{diag}(e^{\lambda A_{ij} b_1}, \dots, e^{\lambda A_{ij} b_d}) U^T$$

$$\mathbb{E} e^{\lambda x_{ij}} = U \text{diag}(\mathbb{E}[e^{\lambda A_{ij} b_1}], \dots, \mathbb{E}[e^{\lambda A_{ij} b_d}]) U^T \rightarrow \mathbb{E}[e^{\lambda A_{ij} b_k}] \leq e^{\frac{\lambda^2 b_k^2 \sigma^2}{2}}$$

$$\rightarrow \mathbb{E}[e^{\lambda x_{ij}}] \leq U \text{diag}(e^{\frac{\lambda^2 b_1^2 \sigma^2}{2}}, \dots, e^{\frac{\lambda^2 b_d^2 \sigma^2}{2}}) U^T = e^{\frac{\lambda^2 \sigma^2 (B_{ij})^2}{2}}$$

Q6) continued: now let's bound the operator norm of A

$$P(\|A\| > t) = P(\|D(A)\| > t) = P(e^{\lambda \|D(A)\|} > e^{\lambda t})$$

$$\leq E[e^{\lambda \|D(A)\|}] e^{-\lambda t} \leq E[\|e^{\lambda D(A)}\|] e^{-\lambda t} \leq E[\text{tr}(e^{\lambda D(A)})] e^{-\lambda t}$$

$$(D(A) = \sum_{i,j} \lambda_{ij}) \rightarrow \text{tr}(\psi_{\lambda_{ij}}(\lambda)) e^{-\lambda t} \leq \text{tr}(e^{\sum \psi_{\lambda_{ij}}(\lambda)}) e^{-\lambda t}$$

$$\leq \text{trace}(e^{\sum \frac{\lambda_{ij}^2 \sigma^2}{2} \beta_{ij}^2}) e^{-\lambda t} \quad \left(e^{\sum \frac{\lambda_{ij}^2 \sigma^2}{2} \beta_{ij}^2} = \begin{pmatrix} e^{\frac{n \lambda^2 \sigma^2}{2}} & & 0 \\ & \ddots & \\ 0 & & e^{\frac{m \lambda^2 \sigma^2}{2}} \end{pmatrix} \right)$$

$$\leq (m+n) e^{\frac{\lambda^2 \sigma^2}{2} (m+n)} e^{-\lambda t} \quad \lambda = \frac{t}{(m+n) \sigma^2} \quad (m+n) e^{-\frac{t^2}{2(m+n) \sigma^2}}$$

$$\rightarrow P(\|A\| > C \sigma (\sqrt{m} + \sqrt{n} + t)) \leq e^{-t^2} ?$$

turns out the above bound doesn't work for the order

that we want, it can yield an order of: $\sqrt{d} \sqrt{\log d}$

since we want an order of $\sqrt{d}$, we'll have to take

an alternative approach, similar to Q3 of pset 2.

"the ϵ -net argument"

Q6) Lemma: $A \in \mathbb{R}^{n \times n}$ $\mathcal{N}: \varepsilon$ -net on $\mathcal{S}^{n-1}$, $\mathcal{M}: \varepsilon$ -net on $\mathcal{S}^{m-1}$

$$\sup_{\substack{x \in \mathcal{N} \\ y \in \mathcal{M}}} |\langle Ax, y \rangle| \leq \|A\| \leq \frac{1}{1-2\varepsilon} \sup_{\substack{x \in \mathcal{N} \\ y \in \mathcal{M}}} |\langle Ax, y \rangle|$$

Since the left side is trivial, we'll prove the right side:

$$|x_1^T A^T y_1 - x_2^T A^T y_2| \leq |(x_1^T - x_2^T) A^T y_1| + |x_2^T A^T (y_1 - y_2)| \leq 2 \|A\| \varepsilon$$

(take x_i, y_i to be the corresponding points in the ε -net)

$$\rightarrow \|A\| \leq 2\varepsilon \|A\| + \sup_{\substack{x \in \mathcal{N} \\ y \in \mathcal{M}}} \langle Ax, y \rangle \rightarrow \|A\| \leq \frac{\sup(\dots)}{1-2\varepsilon}$$

Note that $\mathcal{N}(B_2^n, \varepsilon) \leq (\frac{2}{\varepsilon} + 1)^n$

$$\langle Ax, y \rangle = \sum_{i,j} A_{ij} x_i y_j \rightarrow \|\langle Ax, y \rangle\|_{\ell_2}^2 = \sum_{i,j} \|A_{ij}\|_{\ell_2}^2 x_i^2 y_j^2$$

$$= \sigma^2 \sum x_i^2 \sum y_j^2 \leq \sigma^2$$

$$\text{therefore: } \mathcal{P}(\langle Ax, y \rangle > t) \leq e^{-\frac{t^2}{2\sigma^2}}$$

$$\text{set } \varepsilon = \frac{1}{4} \rightarrow \mathcal{P}(\max_{x,y} \langle Ax, y \rangle > t) \leq |\mathcal{M}| |\mathcal{N}| e^{-\frac{t^2}{2\sigma^2}}$$

$$\leq 9^{n+m} e^{-\frac{t^2}{2\sigma^2}} \quad (\text{set } t' = C\sigma^2(\sqrt{n} + \sqrt{m} + t))$$

$$\leq e^{-t'^2} \quad (\text{for, because } t'^2 \geq C^2 \sigma^2 (n+m+t^2) \text{ for big enough } C)$$

Q6) continued:

Because of the lemma that we proved, for $\varepsilon = \frac{1}{4}$ we have

$$\|A\| \leq 2 \sup_{u,v} \langle A_{u,v} \rangle \rightarrow P(\|A\| > 2\alpha) \leq P(\sup \langle A_{u,v} \rangle > \alpha)$$

$$\rightarrow P(\|A\| > C' \sigma(\sqrt{m} + \sqrt{n} + t)) \leq e^{-t^2}$$

$$E\|A\| = \int_0^\infty P(\|A\| > t) dt \leq \int_{C' \sigma(\sqrt{m} + \sqrt{n})}^\infty \max\{e^{-t^2}, 1\} dt + \int_0^{C' \sigma(\sqrt{m} + \sqrt{n})} e^{-t^2} dt$$

$$\rightarrow t' = C' \sigma(\sqrt{m} + \sqrt{n} + t')$$

$$= C' \sigma(\sqrt{m} + \sqrt{n}) + C' \sigma' \frac{\sqrt{\pi}}{2} \lesssim \sigma(\sqrt{m} + \sqrt{n})$$